Automated Logical Deciders

Certificate-Indexed Halting

Multiagent proof search with a shared verified lemma pool

A_+: prove c

A_-: prove $\neg c$

$A_\perp$: prove contradiction

A_{ind}: certify independence

Brian Tenneson

Research supplement, August 2026

The old independence problem

Run one ATP on c and another on $\neg c$. If one succeeds, there is evidence. If neither succeeds, nothing follows.

Two non-halting searches do not prove independence.

They may indicate:

- genuine independence;
- inadequate search or training;
- a proof beyond the budget;
- memory exhaustion or a software fault.

The key distinction is

$$\boxed{\text{program stop} \neq \text{search exhaustion} \neq \text{logical settlement.}}$$

A timeout must remain **UNKNOWN**. It cannot become **REFUTED** or **INDEPENDENT** merely because the computer stopped.

The change: search for positive certificates of non-derivability rather than interpreting silence.

How independence becomes detectable

An independence agent works in a declared metatheory $\mathcal{M}$ and searches for finite, checkable evidence that

$$T \nvdash c \qquad \text{and} \qquad T \nvdash \neg c.$$

Possible certificate pairs include:

- two countermodels: one of $T + \neg c$, one of $T + c$;
- finite exhaustive saturation excluding both sides;
- two preserved invariants;
- consistency certificates for $T + c$ and $T + \neg c$;
- a formal metaproof, such as a checked forcing or conservation argument.

Independence is detected by a successful certificate search, not by two failed proof searches.

The conclusion is always relative to the frozen object theory, proof calculus, and metatheory.

Four cooperating agents

Proponent A_+. Searches for a verifier-accepted proof of c.
Opponent A_-. Searches for a verifier-accepted proof of $\neg c$.
Inconsistency agent $A_\perp$. Searches directly for $\perp$ or combines proofs of c and $\neg c$.
Independence agent A_{ind}. Searches for two checked nonderivability certificates.
Possible outcomes:

DERIVABLE	$T \vdash c$
REFUTABLE	$T \vdash \neg c$
BOTH	both sides derive
INCONSISTENT	$T \vdash \perp$
INDEPENDENT	neither side derives
UNKNOWN	no conclusive certificate

A status is conclusive only when its own verifier accepts the appropriate certificate.

The shared lemma pool

At every round, every agent reads the current verified pool, performs local search, submits proposed additions to independent checkers, and receives every accepted contribution in the next round.
The pool is typed and provenance-aware:

$$\mathbb{L}_r = \mathcal{L}_r^{\mathcal{F}} \sqcup \mathcal{K}_r^{\mathcal{M}}.$$

An object record

$$(\mathrm{obj}, \Delta, \ell, \pi)$$

certifies $T \cup \Delta \vdash \ell$. It may be reused only in a context that contains or discharges its support Δ. A metatheoretic record carries its own checker proof in $\mathcal{M}$.

Every agent can add to and use the verified pool in every round.

When every member of Δ is later proved from T, the record is promoted to a base lemma. Provenance therefore permits aggressive cross-agent reuse without leaking branch assumptions. Reuse can alter search cost and order, but not theoremhood.

Halting rules and future ML guidance

Licensed conclusive halting:

- checked proof of c;
- checked proof of $\neg c$;
- checked contradiction;
- checked independence certificate;
- certified finite closure excluding a target.

Operational halting only:

- time, expansion, or memory budget exhausted;
- no promising branch remains under a heuristic;
- agent crash or interrupted run.

These produce **UNKNOWN**.

Machine learning later

ML may rank agents, goals, inferences, model sizes, invariant templates, and pool lemmas. It may estimate which status is closest and allocate compute accordingly.
Soundness remains unchanged because ML only proposes. Independent verifiers decide what enters the pool and which status is accepted. A protected fair exploration channel preserves eventual certificate discovery despite imperfect rankings.

Originality assessment. Cooperative provers, blackboards, lemma reuse, model-based nonconsequence, and ML guidance have precedents. The candidate new contribution is their certificate-indexed ALD synthesis with status-specific halting, assumption-provenance, and universal per-round pool access. See the full paper for sources and qualifications.

Search broadly. Share everything verified.
Settle only by certificate.